

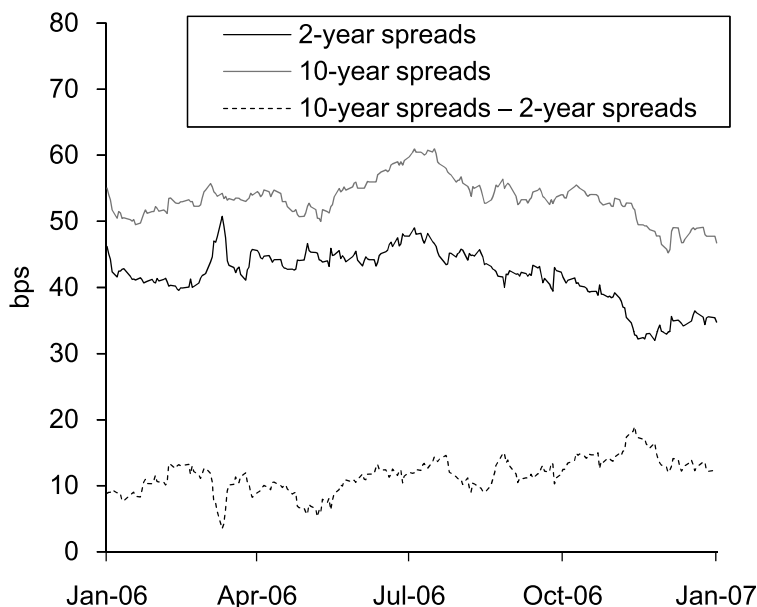


Figure 8.17: The Spread of Spreads Between 10-Year Swap Spreads and 2-Year Swap Spreads

as well as the difference between them. People often look at these spread relationships and look for trading opportunities. Sometimes people put these trades on to take a view on either the credit curve (using swap

coupon of 5.47% in a year exactly offsets the negative carry in the first year. We know this from Equation 8.1.

When we buy \$100mm 4-year Treasuries with a coupon of 5.19% at par and finance the position for a year at 4.80%, we earn carry of \$390,000 for the year (the difference between the coupon interest of \$5,190,000 and the financing costs of \$4,800,000). By adapting Equation 8.1 for Treasuries, we can lock in a *forward price* of \$100mm – \$390,000 = \$99.61mm at which we can sell these Treasuries in a year. The *forward yield* of the Treasury, y , corresponding to this price solves

$$\frac{\$5,190,000}{1+y} + \frac{\$5,190,000}{(1+y)^2} + \frac{\$105,190,000}{(1+y)^3} = \$99.61\text{mm},$$

or $y = 5.334\%$. Note that in a year from now the 4-year Treasuries that we purchased will have three years until maturity, and a new 3-year Treasury will be auctioned. Swap spreads are typically quoted over the on-the-run Treasury of the appropriate maturity. We will assume (somewhat artificially) that there is no premium for on-the-run Treasuries, and thus the newly issued 3-year will also have a yield of 5.334%. Thus, the *forward swap spread* must be equal to $5.644\% - 5.334\% = 31.0$ bps. Thus, we needed spreads to widen out $31.0 - 28.0 = 3.0$ bps from the spot level of spreads in order to break even.

spreads as a proxy for credit spreads), forward GC-LIBOR expectations, or a view on future Treasury supply. Other times people are interested in putting on these trades when the spread of spreads gets out of whack, meaning it deviates from its long-run mean significantly. If an investor believed that the spread of spreads between 10s and 2s was too narrow, he might look to put on a trade that will profit if the spread of spreads widens back out. This would amount to simultaneously buying 10-year spreads and selling 2-year spreads, as discussed earlier. Typically, one would duration-weight each leg of the trade such that the value of a bp move in each spread would have the same dollar value.

8.3.3 Conditional Spread Trades

As mentioned in the introduction to this chapter, sometimes people use options to exploit opportunities they find in the outright market. This typically involves buying one or more options that are considered “cheap” and simultaneously selling one or more options that are considered “rich.” Buying and selling a combination of options may put an investor in a trade that, at option expiry, has “better levels” than could otherwise be obtained in the outright market. But this is not going to leave the investor in a trade that exhibits the same exact view as the original trade that didn’t employ options, and it will also introduce a new set of risks that should be considered.

Example

Suppose that 10-year Treasury yields are at 5.66%, 10-year swap rates are at 6.05%, and thus 10-year swap spreads are at 39 bps. Consider the following trade:

- Buy \$100mm of a 6 month into 10 year payer swaption struck at 6.15% for 62 bps.
- Sell \$100mm of a 6-month put option on the then-on-the-run 10-year Treasury struck at 5.80% for 66 bps.⁴⁰

There are a few things worth noting about this trade. First, right off the bat an investor who puts this trade on puts \$40,000 in his pocket. He sells one option to a dealer for 66 bps and buys the other from the dealer for 62 bps, netting himself 4 bps, which, on a notional of \$100mm, is \$40,000. We should not refer to this as profit; this is simply the net premium received for entering the trade. In fact, the moment that the investor takes in this

⁴⁰A put option on a particular Treasury is an option that gives the holder the right to sell the Treasury at a prespecified price. We could of course regard this equivalently

\$40,000, the net mark-to-market on the two options is $-\$40,000$, assuming the overall trade is an $NPV = 0$ transaction.

In general, people tend to favor such option trades that pay them an upfront and shy away from ones that require they shell out cash to enter. A trade that pays out an upfront is offering no free lunch. A trade that is structured to pay an upfront payment today will have a negative mark-to-market of equal magnitude once the payment has been made, such that the trade overall is an $NPV = 0$ transaction.

In order for both options to finish in-the-money, the market needs to sell off from its current levels. If both options finish in-the-money, then (1) the investor will exercise his payer swaption in six months and pay fixed at a rate of 6.15% for 10 years to the dealer, and (2) the put option will be exercised against him, and the dealer on exercise will sell on-the-run 10-year Treasuries to the investor at a yield of 5.80%. That is, the investor will buy 10-year on-the-run Treasuries at a yield of 5.80% in the event the dealer exercises.

So if both options are exercised, the investor will be paying fixed and will be buying Treasuries—just like the spread widening trade that we talked about earlier. Remember, this will only happen conditional on both options finishing in-the-money, which happens when rates are high enough. Thus, conditional on rates being high enough at maturity, the investor will be in a 10-year swap spread widener at a level of $6.15\% - 5.80\% = 35$ bps. Note that the entry level of 35 bps for the widener compares favorably to the spot level of 10-year swap spreads that was stated to be 39 bps.⁴¹ A lower entry point is beneficial on a swap spread widener trade, since the trade profits by the amount that spreads finish above the entry point.

as an option to sell the Treasury at a particular yield, since for every Treasury price there corresponds a unique yield. At expiry, the put option will be exercised if the price of the Treasury is below the strike, or equivalently if the yield of the Treasury is above the yield strike. Let's consider now the concept of a put option on a then-on-the-run Treasury. Recall that the U.S. Treasury issues new Treasury securities on a regular cycle. The 10-year Treasury that is the on-the-run today will not be the on-the-run in, say, six months. A put option on the then-on-the-run 10-year Treasury that expires in six months is an option on a security that does not yet exist. This put option is written on whatever security happens to be the on-the-run 10-year at the time of expiration six months from now. Although the underlying itself does not exist at this point in time, dealers who make markets in these types of options are willing to make certain assumptions as to how this security will trade in relation to existing securities so as to be comfortable putting a price on such options. Since the underlying security does not yet exist, it does not yet have a coupon, and hence we cannot speak about a dollar price of the security. Therefore, options on then-on-the-run Treasuries have a strike that is quoted in yield terms.

⁴¹Comparing the conditional entry point on this trade, which is 35 bps, to the spot level of 10-year spreads is an apples to oranges comparison. It would be more meaningful to compare this level to where the market implies 10-year swap spreads are six months forward. Such a trade exists in the swap market and is referred to as a *spread lock*, which is a forward on a swap spread.

It is important to recognize that an investor who puts on this conditional spread widener is taking a fundamentally different view (whether he realizes it or not) than an investor who puts on the plain vanilla (unconditional) spread widener we examined earlier. A regular spread widener benefits if spreads widen, no matter what happens to rates. This conditional spread widener benefits at expiry only if spreads widen in conjunction with a market sell-off.

Note that if both options finish in-the-money, this does not guarantee the investor a positive payout at maturity (exclusive of the \$40,000 upfront). If, for example, in six months 10-year swap rates are at 6.20% and 10-year Treasury yields are at 5.90%, the investor will lose money. The payer swaption that he is long is in-the-money 5 bps (running), but the put option he is short is in-the-money 10 bps (running).

Figure 8.18 shows the net payoff at expiration of the options as a function of where rates are. Swap rates and Treasury yields (which differ only by the swap spread) tend to move in the same direction. Thus, an investor who puts

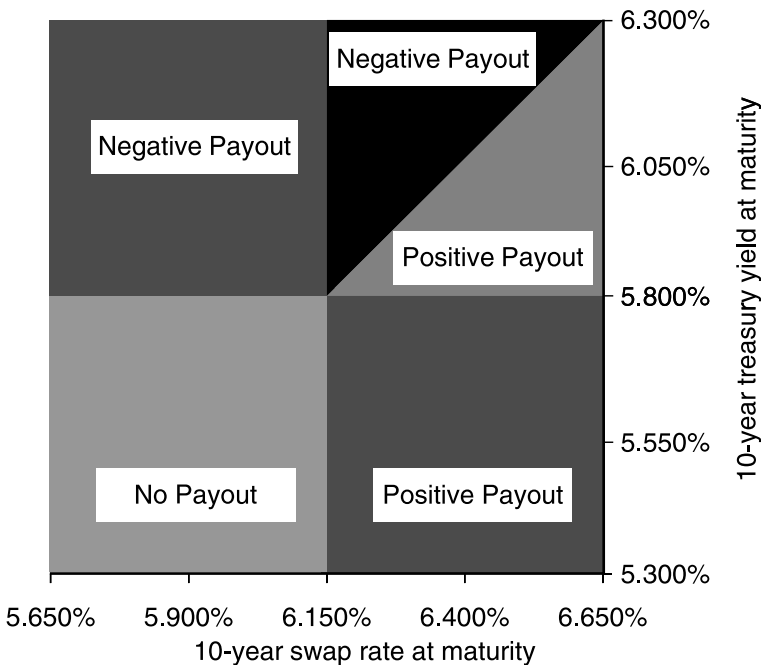


Figure 8.18: Net Payout at Maturity of Option Position (Exclusive of Upfront Fee) as a Function of Treasury Yields and Swap Rates at Expiration

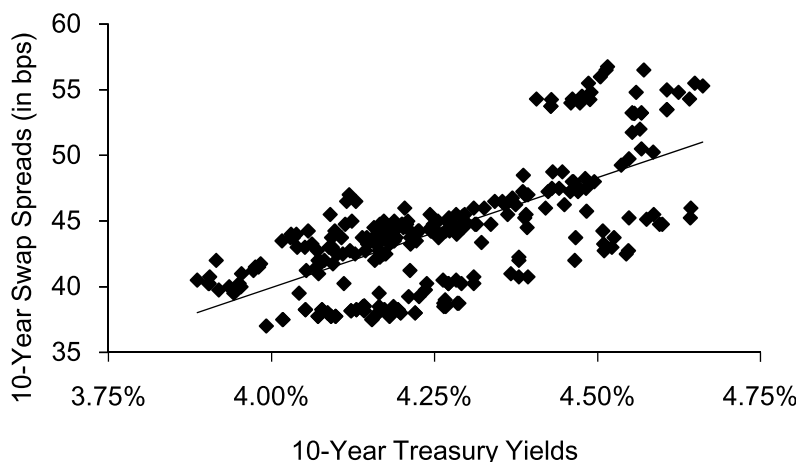


Figure 8.19: The Relationship Between 10-Year Treasury Yields and 10-Year Swap Spreads in the Second Half of 2005

this trade on will tend to find himself in either the lower left-hand quadrant or the upper right-hand quadrant at expiry. The lower left-hand quadrant corresponds to the case in which the market has rallied, and both options finish out-of-the-money. In this case, the investor earned the \$40,000 upfront in the trade and had no further payout. The upper right-hand quadrant corresponds to the case in which the market has sold off, and both options are in-the-money. In this case, the payout at expiry can be either positive or negative. If 10-year spreads are tighter than 35 in this scenario, the payout will be negative, and otherwise it will be positive. Of course, the investor may find himself in the upper left-hand or lower right-hand quadrants, obtaining a negative payout or positive payout, respectively.

The dynamic of spreads widening in a sell-off is one that many investors have grown accustomed to over the years. For example, as we discussed in Section 2.2.1, during the summer of 2003 the violent sell-off in the market was accompanied by a widening of spreads. Figure 8.19 shows the relationship between Treasury yields and swap spreads during the second half of 2005. As this figure shows, the general trend during this period of time is one of higher Treasury yields corresponding with higher swap spreads.⁴² Scenarios such as these have made many people accustomed to the belief that as the market sells off, swap spreads should widen. This trade positions one for

⁴²This figure also shows, however, that there are times when high Treasury yields are associated with tighter swap spreads.

exactly that; provided that the market sells off enough from current levels and spreads remain wide enough, the trade provides a positive payout at expiry. Of course in the case in which the market sells off and spreads tighten (a dynamic that has been observed at certain times in the past), this trade will have a negative payout at expiry. And if at expiry spreads tighten when the market rallies (another commonly held belief), then one will not be hurt at expiry since both options will finish out-of-the-money. In fact, in this case the investor will keep the \$40,000 that he originally pocketed.⁴³

This trade is referred to in the industry as *puts payers*. One of the drivers of being able to get in at the seemingly great strike differential of 35 bps (as compared to the spot level of 10-year spreads of 39 bps) is the differential in the Treasury volatility one is selling versus the swaption volatility one is buying. Treasury vol and swaption vol do not necessarily move together, and as Treasury vol tends to richen versus swaption vol, such puts payers trades become attractive. Of course, one of the risks an investor bears on entering this trade is the exposure to both volatilities. If after this trade is put on, implied volatility on the Treasury option that has been sold increases and implied volatility on the swaption decreases, the investor will face a mark-to-market loss on the trade.

The risks associated with implied volatility in this trade can be a significant factor in its success or failure. Suppose, for instance, that one month after entering into the trade, the payer swaption is in-the-money and the Treasury option is at-the-money. In this case the trade appears to be going well for the investor. Suppose, however, that there is large overseas buying of Treasury options. Implied Treasury vol may spike dramatically. This will hurt the mark-to-market on the Treasury option. Although swaption vol may rise somewhat out of sympathy for the move in Treasury vol, the payer swaption is in-the-money and thus has limited vega exposure.⁴⁴ This adverse move in vol, despite the desired move in spreads in conjunction with a sell-off, may move the mark-to-market on the trade significantly against the investor, causing him to unwind the trade if he hits his predetermined stop-loss.

One final thing to note about this trade: sometimes investors would rather forgo an upfront fee (in this case, the \$40,000) and improve their entry level. Thus, as an alternative to the original terms of the trade, one may be able to put on the following trade:

⁴³Although the investor will have zero payout at expiry in the case in which spreads tighten and the market rallies, the investor may have a substantially negative mark-to-market during the life of the trade.

⁴⁴In markets that are somewhat related, such as the swaption vol and Treasury vol markets, a move in one market due to a particular catalyst may spur movement in the other market. Such a move is referred to as occurring *out of sympathy*.

- Buy \$100mm of a 6 month into 10 year payer swaption struck at 6.14% for 66 bps.
- Sell \$100mm of a 6-month put option on the then-on-the-run 10-year Treasury struck at 5.80% for 66 bps.

In this case, the strike on the payer swaption has been reduced to 6.14%, implying that the conditional entry point on the spread widener is now $6.14\% - 5.80\% = 34$ bps, instead of the 35 bps that was available in the version of the trade with an upfront fee of \$40,000.⁴⁵ Many investors prefer the version with the upfront, because if the market rallies and the options finish up out-of-the-money, they at least get to keep the upfront. But the choice is a matter of investor preference. $\square$

8.4 Asset Swaps Revisited

Recall that in Section 2.5.2 we said that an asset swap referred to the simultaneous purchase of a bond and use of a swap to transform the nature of the asset from either fixed to floating, or vice versa. Most of the time the bond in question is a fixed rate bond, and the buyer of the asset swap buys the bond and pays fixed against it. Asset swaps represent a large segment of the types of trades that relative value investors put on in the swap market. The most commonly asset swapped securities are Treasuries and agency debentures. Investors both buy and sell asset swaps, looking to buy asset swaps when they think they are cheap and sell them when they are rich.

There are a number of different methodologies that are employed to put on asset swaps. The classical transaction (which is seldom used anymore for reasons we shall discuss) is referred to as a *par asset swap*. In order to buy a par asset swap, an investor

1. Buys a fixed rate bond;

⁴⁵As a handcheck on the strike differential obtainable with no upfront fee, consider the following. By setting

$$\begin{aligned} \text{PMT} &= 1.0 \text{ bp}/2 = 0.5 \text{ bp} \\ n &= 20 \\ i &= 6.14\%/2, \end{aligned}$$

we find that a 1 bp change in coupon on this 10-year swap is worth approximately \$74,000. An at-the-money option has a delta around 0.5, and thus a 1 bp change in the strike is worth around \$37,000. Thus, forgoing the fee of \$40,000 implies that we should be able to improve the strike differential by approximately 1 bp.

2. Pays fixed in a swap (in which the notional matches the face amount of the bond) where the fixed payments *exactly* match the coupon payments of the bond in both timing and magnitude;
3. Makes or receives an upfront payment equal to the *dirty price* minus par (if the dirty price minus par is positive, then the investor receives this amount; otherwise, he pays it as an upfront);⁴⁶
4. Receives a floating rate (typically 3-month LIBOR) plus an agreed-upon spread to the floating rate on a notional that is again equal to the face amount of the bond.

The agreed-upon spread is referred to as the *asset swap spread*. The asset swap spread is set such that the swap is an $NPV = 0$ transaction. It is important to note that an investor who buys a bond and asset swaps it still has credit exposure to the issuer. If the credit should deteriorate, this will hurt the investor.⁴⁷

Example

Let's look at an asset swap in the context of our stylized example, and suppose that the swap curve is as given in Table 8.1. Consider a 4-year agency bond with a 6% coupon (annual bond) that is trading at a yield of $T_4 + 16$. The bond has no accrued interest, and it can be shown that its (dirty) price is 102.286252.⁴⁸ If an investor buys \$100mm face of the bond and asset swaps it with a swap dealer, the dealer will be receiving the fixed 6% coupons on the bond, paying an upfront fee of \$2,286,252 (the difference between the dirty price and par) and will be receiving LIBOR plus a spread. This transaction is depicted in Figure 8.20. In the top half of Figure 8.21, it is shown that the spread to 1-year LIBOR that makes this an $NPV = 0$ transaction is

⁴⁶The dirty price of a bond refers to the full-invoice price. It is the quoted price plus accrued interest.

⁴⁷In the event of default by the issuer of the bond, the investor who entered into the asset swap will still have to make fixed payments versus LIBOR in the swap he has entered into, unless a specific clause in the terms of the trade was negotiated in which case payments in the swap would cease upon default by the issuer of the reference credit. Such clauses are rare and nontrivial to negotiate and price and would potentially impact the asset swap spread in the trade.

⁴⁸From Table 8.1, we have that T_4 is 5.19%, and thus the yield on the bond is $5.19\% + 0.16\% = 5.35\%$. The price of the bond is

$$102.286252 = \frac{6}{1.0535} + \frac{6}{(1.0535)^2} + \frac{6}{(1.0535)^3} + \frac{106}{(1.0535)^4}.$$

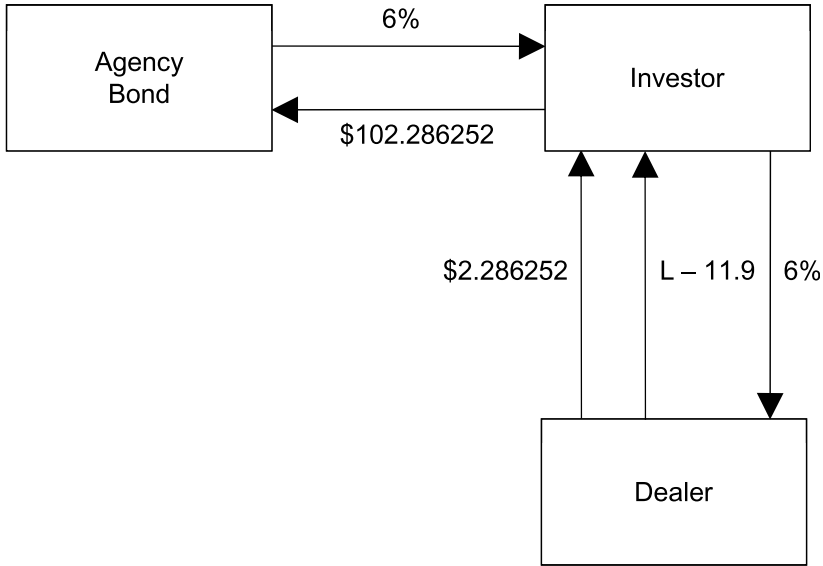


Figure 8.20: Asset Swapping a 4-Year Agency Security

–11.9 bps. We say that the asset swap spread for this agency bond is –11.9 bps, or that the bond asset swaps to –11.9 bps.⁴⁹ □

The million-dollar question with asset swaps is what spread to LIBOR the investor is able to get. If the investor is buying the asset swap, then he is looking for a large spread; the larger the spread, the “cheaper” the asset swap, and the smaller the spread, the “richer” the asset. If he is selling the asset swap, then he wants the spread to be as small as possible.⁵⁰

⁴⁹Recall that the agency bond yields $T_4 + 16$, and the swap rate is equal to $T_4 + 28$. Notice that the asset swap spread is approximately equal to the difference between the agency spread and the swap spread. That is,

$$-11.9\text{bps} \approx 16\text{bps} - 28\text{bps}.$$

As we will see, this is not a coincidence.

⁵⁰It should not be surprising that agency bonds sometimes have asset swap spreads that are negative. Many market participants think of LIBOR as representing the credit quality of a AA bank. The agencies have (at times) been regarded as having better than AA credit quality, which would suggest that they might asset swap through LIBOR, or “sub-LIBOR” as people sometimes say.

Time	zc	Forward	Fixed	PV Fixed	L + Spread	PV(L + Spread)
1	5.000%	5.000%	6,000,000	5,714,286	(4,880,689)	(4,648,275)
2	5.113%	5.226%	6,000,000	5,430,501	(5,106,437)	(4,621,752)
3	5.312%	5.712%	6,000,000	5,137,062	(5,592,891)	(4,788,505)
4	5.494%	6.041%	6,000,000	4,844,394	(5,922,053)	(4,781,460)
21,126,244						(18,839,992)
						upfront (2,286,252)
		Fixed rate	6.000%	NPV	(0)	
		Notional	100,000,000	Asset swap spread		-0.119%

Computing LIBOR Price of Bond				Computing Dirty Price of Bond		
Time	zc	CF	PV(Fixed Bond)	Yield	CF	Dirty Price
1	5.000%	6,000,000	5,714,286	5.350%	6,000,000	5,695,301
2	5.113%	6,000,000	5,430,501	5.350%	6,000,000	5,406,076
3	5.312%	6,000,000	5,137,062	5.350%	6,000,000	5,131,539
4	5.494%	106,000,000	85,584,302	5.350%	106,000,000	86,053,335
101,866,152						102,286,252
		Notional	100,000,000	PV(Spread)	(420,100)	
				(in bps)	(42.0)	
Time	zc	bps Running	CF	PV(CF)		
1	5.000%	-11.9	(119,311)	(113,630)		
2	5.113%	-11.9	(119,311)	(107,987)		
3	5.312%	-11.9	(119,311)	(102,152)		
4	5.494%	-11.9	(119,311)	(96,332)		
				(420,100)		

Figure 8.21: Determining the Asset Swap Spread of the 4-Year Agency (*top half*) and Showing That the Present Value of the Asset Swap Spread Is Equal to the Difference Between the LIBOR Price of the Bond and Its Dirty Price (*bottom half*)